

Technology Stack

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

Layer	Technology	Purpose
frontend	HTML5	Structure of web pages
Frontend	CSS3	Styling and responsive design
Frontend	JavaScript	Client-side interactivity
Frontend Framework	Bootstrap	Responsive UI components
Backend	PHP	Server-side application logic
Database	MySQL	Store user, flood, and alert data
Web Server	Apache (XAMPP/WAMP)	Hosts the web application
Development Environment	Visual Studio Code	Code development and editing
Version Control	Git	Source code management
Repository Hosting	GitHub	Store and collaborate on project code
Maps API	Google Maps API	Display flood locations on maps
Notification Service	Email (SMTP)	Send alerts and notifications
Authentication	Session-Based Login	Secure user authentication
Data Exchange	JSON	Transfer data between frontend and backend
Browser Support	Google Chrome, Microsoft Edge, Firefox	Run the web application
Operating System	Windows 10/11	Development and deployment environment

Technology Stack Summary

Component	Technology Used
Frontend	HTML5, CSS3, JavaScript, Bootstrap
Backend	PHP
Database	MySQL
Web Server	Apache (XAMPP/WAMP)
IDE	Visual Studio Code
Version Control	Git, GitHub
APIs	Google Maps API, SMTP
Data Format	JSON
Authentication	Session-Based Login

